

The big picture

Guiding question(s)

How does the classification of organic molecules help us to predict their properties?

Keep the guiding question in mind as you learn the science in this subtopic. You will be ready to answer it at the end of this subtopic. The guiding questions require you to pull together your knowledge and skills from different sections, to see the bigger picture and to build your conceptual understanding.

Organic chemistry is a branch of chemistry that focuses on molecular compounds containing carbon (not to be confused with organic farming, which is farming without the use of artificial pesticides and fertilisers). The term 'organic' originally referred to compounds derived from living organisms. This originated from the concept of 'vitalism', a theory that organic compounds could only be formed by living organisms. This theory was disproved in 1828 when Frederick Wohler, a German chemist, successfully created urea ($\text{CO}(\text{NH}_2)_2$) in the laboratory. Up until this point, it was thought that urea, which is found in urine, could only be produced in the kidneys of living organisms.

Fast forward to the modern day, and the number of organic compounds is staggering. It is estimated that there are more than 10 000 000 (ten million) different organic compounds. They are used extensively in our daily lives from the fuels that power our vehicles to the plastics that make up our devices. **Figure 1** shows some common everyday substances made from organic compounds.

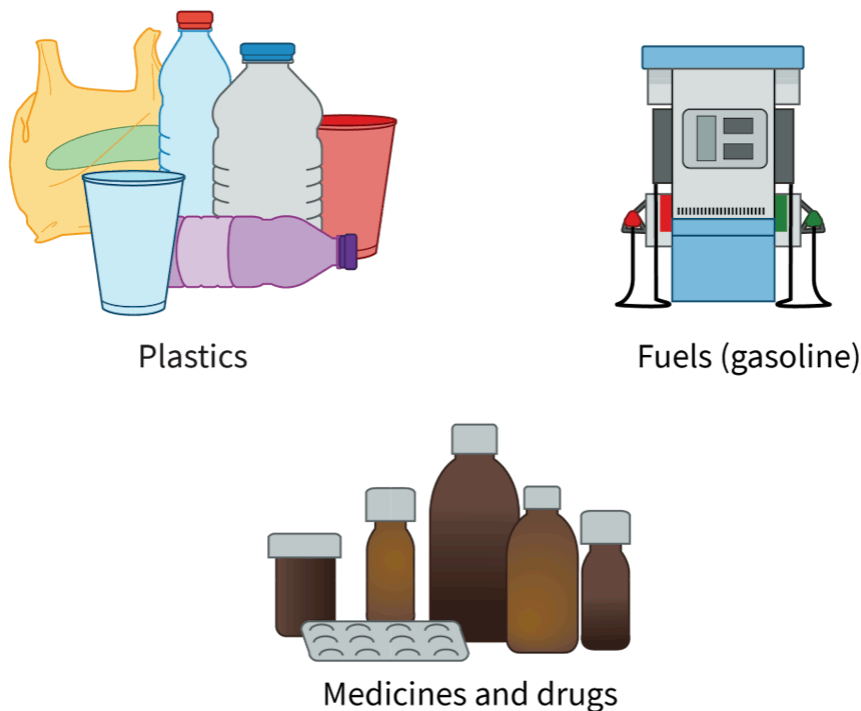


Figure 1. Organic compounds are found in a variety of everyday items.

Having such a wide range of different compounds requires a comprehensive system of organisation and naming. This naming system is known as IUPAC nomenclature, which will be covered in more detail in sections S3.2.5a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nomenclature-of-straight-chain-organic-id-45147/>) and S3.2.5b (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/nomenclature-of-branchedchain-organic-compounds-id-45664/>). IUPAC is the International Union of Pure and Applied Chemistry, an organisation that is responsible for devising the rules for naming organic compounds. As you will learn in this subtopic, organic compounds are classified into homologous series (sections S3.2.3–4 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/>))) that have similar chemical and physical properties. This enables us to predict the properties of an organic compound based on the series to which it belongs.

Prior learning

Before you study this subtopic make sure that you understand the following:

- Single, double and triple bonds involve one, two and three shared pairs of electrons respectively (see section S2.2.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/multiple-covalent-bonds-id-45128/>))).

- The valence shell electron pair repulsion (VSEPR) model enables the shapes of molecules to be predicted from the repulsion of electron domains around a central atom (see [section S2.2.4](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/>)).
- The nature of the force that exists between molecules is determined by the size and polarity of the molecules. Intermolecular forces include London (dispersion), dipole-induced dipole, dipole—dipole and hydrogen bonding (see [section S2.2.8a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/intermolecular-forces-id-44359/)  (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/intermolecular-forces-id-44359/>)).

Practical skills

Once you have completed this subtopic, practise building molecular models by going to [Practical 4: Organic model building](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/organic-model-building-id-46718/). (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/organic-model-building-id-46718/>).